



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St., Lagao, General Santos City

Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 1 of 6
Document Code: WI-01-MIC-02	HETEROTROPHIC PLATE COUNT IN WATER AND ICE		

1. OBJECTIVE

To determine the number of viable microorganisms in water and ice.

2. SCOPE

This instruction is applicable in the determination of total bacterial count in water and ice used in pre-processing and processing plants.

3. RESPONSIBILITY

The analyst performing the test should ensure that the procedure is followed accordingly.

4. APPARATUS/EQUIPMENT/CONSUMABLES

- 4.1 Alcohol lamps / Fireboy
- 4.2 Alcohol (70% solution isopropyl alcohol)
- 4.3 Autoclave
- 4.4 Autoclave Indicator Tape
- 4.5 Top loading Balance
- 4.6 Aluminum Foil
- 4.7 Tissue paper or paper towel
- 4.8 Spatula
- 4.9 Graduated cylinder (100 ml, Pyrex)
- 4.10 Media bottles (500 ml, Pyrex)
- 4.11 Dilution bottles with screw caps (160 ml, Pyrex)
- 4.12 Sterile Pipettes (10 ml)
- 4.13 Sterile Pipettor
- 4.14 Sterile Pipette tips (100 – 1000 µl)
- 4.15 Sterile Petri dishes with cover (100x15mm, Pyrex)
- 4.16 Incubator (maintained at 35 ± 1°C)
- 4.17 pH meter
- 4.18 Quebec type colony counter
- 4.19 Glass-marking pen
- 4.20 Biosafety Cabinet

5. CULTURE MEDIA

5.1 All information regarding media preparation is placed in LF-W04-01, Reagent Preparation Logbook.

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5.2 Plate Count Agar

Suspend 23.5 g of the medium in one liter of distilled water. Mix thoroughly. Heat with frequent agitation and boil for 1 minute to completely dissolve the powder. Sterilize 15 min at 121°C. Final pH, 7.0 ± 0.2. If agar solidifies, melt completely and temper at 45°C prior to use.

5.3 Butterfield's Phosphate-Buffered Dilution Water

Suspend 34 g of KH_2PO_4 to 500 liter of distilled water. Adjust pH to 7.2 with 1 N NaOH. Bring volume to 1 liter with distilled water. Sterilize 15 min at 121°C. Store in a refrigerator at 4°C.

Dilution Blanks

Take 1.25 ml of above stock solution and bring volume to 1 liter with distilled water. Dispense into bottles to 90 or 99 ± 1 ml. Sterilize 15 min at 121°C.

6. PROCEDURE

❖ Conventional Plate Count Method

- 6.1 Clean working area with alcohol. If procedure is to be done inside a biosafety cabinet, turn on UV light for at least 30 minutes to disinfect the working area.
- 6.2 Transfer materials inside the hood and arrange them accordingly.
- 6.3 Label dilution bottles and Petri dishes.
- 6.4 Shake the sample well 25 times in 30 cm arc within 7 s. The sampling bottle should have enough air space for effective shaking. If the bottle is completely full, aseptically pour out 20-30 ml of water, replace cap and shake.
- 6.5 Using separate sterile pipets, prepare decimal dilutions of 10^{-1} and 10^{-2} by transferring 10 ml of sample to 90 ml of diluent. Avoid sampling foam. Shake all dilutions 25 times in 30 cm (1ft) arc within 7 s.
- 6.6 The series of dilutions 10^{-1} and 10^{-2} is now complete.
- 6.7 Pipet 1 ml of each dilution (10^{-2} , 10^{-1} , undiluted sample) into separate, duplicate, appropriately marked Petri dishes. Reshake dilution bottle 25 times in 30 cm arc within 7 s if it stands more than 3 min before it is pipetted into Petri dish.
- 6.8 Add 12-15 ml plate count agar (cooled to $45 \pm 1^\circ\text{C}$) to each plate within 15 min of original dilution. Mix sample dilutions and agar medium thoroughly and uniformly by alternate rotation and back-and-forth motion of plates on a flat level surface. Let agar solidify.

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6.9 Pour agar onto 2 sterile Petri dishes. Open one plate labelled as “air control” and cover after 15 minutes of exposure. Label unexposed plate as “medium control”.

6.10 Pipet 1 ml of sterile diluent into a Petri dish labelled as “diluent control” and pour agar.

6.11 Invert plates and incubate at 35°C for 48 ± 2 h.

6.12 Count the number of colonies per plate. Record the results per dilution of the plate counted and report APC as CFU per ml.

CALCULATION:

APC = total number of colonies or average number of colonies (if duplicate plates have the same dilution) per plate times the reciprocal of dilution used.

7. CALCULATIONS AND RECORDING COUNTS

To avoid creating a fictitious impression and accuracy when computing APC, report only the first two significant digits. Round off to two significant figures only at the time of conversion to SPC (Standard Plate Count). Round by raising the second digit to the next highest number when the third digit is 6, 7, 8 or 9 and use zero for each successive digit toward the right from the second digit. Round down when the third digit is 1, 2, 3 or 4. When the third digit is 5, round up when the second digit is odd and round down when the second digit is even.

Examples	
Calculated Count	APC
12,700	13,000
12,400	12,000
15,500	16,000
14,500	14,000

7.1 For plates with 25-250 CFU

Formula:

$$N = \frac{\Sigma C}{(1 \times n_1) + (0.1 \times n_2)} d$$

Calculate the APC as follows:

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Where N = Number of colonies per ml or g of sample
 ΣC = Sum of all colonies on all plates counted
 n_1 = no. of plates with 25-250 colonies in lowest dilution
 n_2 = no. of plates with 25-250 colonies in highest dilution
d = dilution from which the first counts were obtained

Example:

1:100	1:1000
232, 244	33, 28

$$N = \frac{(232+244+33+28)}{[(1 \times 2) + (0.1 \times 20)]10^{-2}}$$
$$= 24,409 \approx 24,000$$

When counts of duplicate plates fall within and without the 25-250 colony range, use only those counts that fall within this range.

7.2 For all plates with fewer than 25 CFU.

When plates from both dilutions yield fewer than 25 CFU each, record actual plate count but record the count (EAPC – Estimated Aerobic Plate Count) as less than $25 \times 1/d$ when d is the dilution from which the first counts were obtained.

Example:

Colonies		
1:100	1:1000	EAPC/ml (g)
18	2	<2500
0	0	<2500

7.3 For all plates with more than 250 CFU.

When plates from both 2 dilutions yield more than 250 CFU each (but fewer than 100/cm²), estimate the aerobic counts from the plates (EAPC) nearest 250 and multiply by the dilution.

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Example:

Colonies		
1:100	1:1000	EAPC/ml (g)
TNTC (too numerous to count)	640	640,000

7.4 For all plates with spreaders and/or laboratory accident, report respectively as Spreader (SPR), or Laboratory Accident (LA).

7.5 For all plates with more than an average of 100 CFU per sq. cm. Estimate the APC as greater than 100 times the highest dilution plated, times the area of the plate.

Example:

Colonies / Dilution		
1:100	1:1000	EAPC/ml (g)
TNTC	7,150 ^(a)	<6,500,000 EAPC ^(b)
TNTC	6490	<5,900,000 EAPC

^a Based on plate area of 65cm²

^b EAPC, estimated APC

8. REFERENCES

Bacteriological Analytical Manual, Edition 8, Revision A, 1998. Chapter 3. Authors: Larry Maturin and James T. Peeler

9. DOCUMENTS

LF W01-01, Reagent and Media Preparation Logbook

LF W01-MIC-02, Analyst Worksheet for Bacteriological Examination of Water and Ice

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